

Jayden (Jiafeng) Yu

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TECHNICAL SKILLS

Languages: Python, C/C++, Rust, JavaScript/TypeScript, KDB/SQL, Java/Kotlin, Lisp, Bash, Lua, MIPS/ARM, R
Tools: PyTorch, TensorFlow, NumPy, Pandas, Node.js, Vue.js, MongoDB, Redis, Git, Linux/FreeBSD, LLVM, AWS
Interested in: OCaml, Automated Theorem Proving, Quantum Computing, Pen.Testing, Gentoo/BlackArch Linux
Communication: English, French, Mandarin Chinese, Spanish

EXPERIENCE

Conduit Commerce

New York City, New York

Software Engineer Intern | Python

June 2024 - Present

- E-Commerce & logistics based Stealth Startup

OncAI

San Francisco, California

Machine Learning Engineer Intern | Python

Jan 2024 - May 2024

- Engineered a **3D UNet Diffusion Model** to render synthetic CT/PET scans for medical **data generation**
- Researched and implemented various **data augmentation algorithms** to enhance **learning robustness**
- Implemented and updated features to the **Valohai MLOps pipeline** to unify interfaces
- Implemented **Python** scripts to automate data extraction from medical records

University of Waterloo

Waterloo, Ontario

Undergraduate Research Assistant | Python

Oct 2023 - Ongoing

- Researching **automated proof generation** using **LLM** and performing **proof verification** with **Coq**

Ledn Inc

Toronto, Ontario

Quantitative Developer Intern | Rust, Python

Sept 2022 - Dec 2022

- Designed, engineered, and implemented from scratch a **Rust** based multi-threaded market data consolidator to decrease processing latency **by over 50%** when compared to a FOSS Python model.
- Engineered the design and implementation of a **ML** quant trading product in **Python** and **Rust**, through researching various limit order models, efficient documentation, and code reviews
- Fine-tuned crypto market data to produce supervised learning datasets using **KDB** and **SQL**
- Managed an **AWS EC2** instance to source, store, and update modeling data and host a KX Dev environment
- Researched, implemented, and performance tested a **LSTM** model in **PyTorch** for price prediction

Ford Motor Company

Waterloo, Ontario

Software Engineer Intern | Python, JavaScript

Jan 2022 - April 2022

- Implemented the **backend development** of a performance metrics website in **JavaScript**
- Engineered **Python** scripts to automate updates to the **data processing pipeline** via **JSON** configuration files

SS&C Technologies

Toronto, Ontario

Full Stack Developer Intern | Java, JavaScript

May 2021 - Sept 2021

- Implemented modules of a **Java** based **Selenium** webpage testing framework from scratch
- Designed and implemented **frontend** components of a revamped legacy product using **Vue.js**

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Mathematics in Honours Computer Science, minor in Pure Mathematics

Sep 2020 - May 2025E

- **Relevant Coursework:** ML, AI, Computer Vision, Algorithms, Data Structures, OS, OOP, Numerical Computation, Compiler Construction, Formal Methods, Computer Architecture, Abstract Algebra, Real Analysis

PROJECTS

Immune System Invaders | JavaScript, HTML, CSS

Mar 2020

Created a web based implementation of the Space Invaders game with a COVID-19 twist in standard **JavaScript**, **HTML**, **CSS** with **OOP** principles, scalability, and maintainability in mind